

MAE 4730
Lecture 32
11/9/18

Today:

• Diads, Lin fins, rotations

Scalar	vector	Tensors
c	$\vec{v}$	$\underline{\underline{R}}, R_{ij} \vec{e}_i \vec{e}_j$
c	$\begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix}$	$\begin{bmatrix} A_{11} & A_{12} & A_{13} \\ A_{21} & \dots & \dots \\ \vdots & \dots & \dots \end{bmatrix}$
c	c_i	A_{ij}

Diads:

$$\vec{a} \vec{b} \quad \text{or} \quad \vec{a} \otimes \vec{b}$$

↑ tensor product ↑

As opposed to scalar product: $\vec{a} \cdot \vec{b} = \text{scalar}$
 vector product: $\vec{a} \times \vec{b} = \text{vector}$
 $\Rightarrow$ tensor product: $\vec{a} \vec{b} = \text{tensor}$

Rules:

- all normal rules of multiplication w/ some exceptions

$$\text{ex) } \vec{a} (\vec{b} + \vec{c}) = \vec{a} \vec{b} + \vec{a} \vec{c}$$

$$c \vec{a} \vec{b} = \vec{a} (c \vec{b}) = \vec{a} \vec{b} c$$

Key things:

$$\begin{aligned} \vec{a} \vec{b} \vec{c} &= \vec{a} (\vec{b} \cdot \vec{c}) \\ \vec{c} \cdot \vec{a} \vec{b} &= (\vec{c} \cdot \vec{a}) \vec{b} \end{aligned}$$

Exceptions:

$$\begin{aligned} \vec{a} \vec{b} &\neq \vec{b} \vec{a} \\ \vec{a} \vec{b} &\neq -\vec{b} \vec{a} \end{aligned}$$

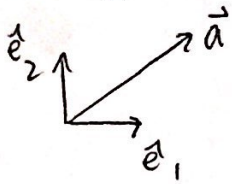
$$\vec{a} \vec{b} = (\vec{b} \vec{a})^T$$

$$\frac{\text{blah}}{\vec{a} \vec{b}} = \text{nonsense}$$

$$\vec{a} \vec{b} \vec{c} = \text{nonsense}$$

$$\text{ex) } \hat{e}_i \hat{e}_i \cdot \vec{a} = \vec{a} \quad \text{--- } a_i \hat{e}_i$$

"A tensor dotted with a vector is a vector"



$$\Rightarrow \hat{e}_i \cdot \hat{e}_i = \underline{\underline{1}} \quad \text{--- identity tensor}$$

$$\underline{\underline{A}} = A_{ij} \hat{e}_i \hat{e}_j$$

$$\text{ex) } \underline{\underline{A}} \cdot \vec{Q} = A_{ij} \hat{e}_i \hat{e}_j \cdot (Q_k \hat{e}_k)$$

pick j or k and go with it

$$= A_{ij} Q_k \hat{e}_i \underbrace{\hat{e}_j \hat{e}_k}_{\delta_{jk}}$$

$$= A_{ij} Q_j \hat{e}_i \quad \delta_{jk} = \begin{cases} 1 & i=k \\ 0 & i \neq k \end{cases}$$

$$[\underline{\underline{A}} \cdot \vec{Q}]_i = A [\vec{Q}]_i$$

$$A_{ij} = \hat{e}_i \cdot \underline{\underline{A}} \cdot \hat{e}_j$$

def

$$A = [\underline{\underline{A}}]_f = \begin{bmatrix} A_{11} & A_{12} & A_{13} \\ A_{21} & & \\ \vdots & & \end{bmatrix}$$

$$\underline{\underline{A}} = \underline{\underline{A}}$$

$$\{A_{ij} \hat{e}_i \hat{e}_j = A_{ij} \hat{e}_i \hat{e}_j\}$$

$$\hat{e}_k \cdot \{ \quad \} \cdot \hat{e}_l \Rightarrow$$

$$\hat{e}_k' \cdot (A_{ij}' \hat{e}_i' \hat{e}_j') \cdot \hat{e}_l' = \hat{e}_k' \cdot (A_{ij} \cdot \hat{e}_i \cdot \hat{e}_j) \cdot \hat{e}_l'$$

$$\underbrace{(\hat{e}_k' \cdot \hat{e}_i')}_{\delta_{ki}} \underbrace{(\hat{e}_j' \cdot \hat{e}_l')}_{\delta_{jl}} A_{ij}' = (\hat{e}_k' \cdot \hat{e}_i) A_{ij} (\hat{e}_j \cdot \hat{e}_l')$$

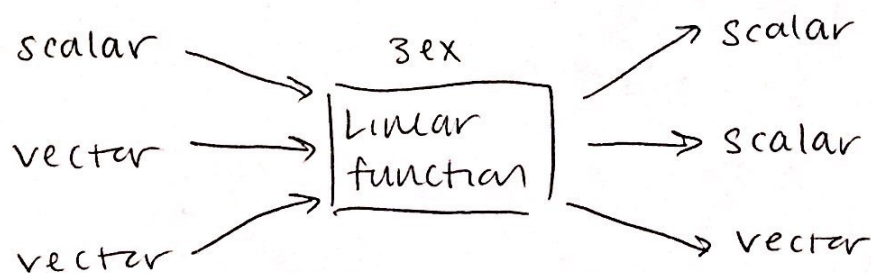
$$A_{kl}' = R_{ik} A_{ij} R_{jl}$$

$$= R_{ki}^T A_{ij} R_{jl}$$

$$[\underline{A}]_{\beta} = R^T [\underline{A}] R$$

$$A = R' A' R$$

Linear Functions:



Linear function: obeys superposition

$$\mathcal{L}(ax + by) = a \mathcal{L}(x) + b \mathcal{L}(y)$$

$$\rightarrow \text{say } \mathcal{L}(1) = 7$$

$$\mathcal{L}(2) = \mathcal{L}(2 \cdot 1) = 2 \mathcal{L}(1) = 14$$

Generally:

$$\mathcal{L}(x) = \mathcal{L}(x \cdot 1) = x \mathcal{L}(1)$$

$$= \mathcal{L}(1) \cdot x$$

$$\boxed{\mathcal{L}(x) = cx} \quad \text{most simple linear function}$$

ex) $\mathcal{L}(\vec{v}) = \text{scalar valued function of a vector}$

$$\mathcal{L}(v_1 \hat{e}_1 + v_2 \hat{e}_2 + v_3 \hat{e}_3) = v_1 \mathcal{L}(\hat{e}_1) + v_2 \mathcal{L}(\hat{e}_2) + v_3 \mathcal{L}(\hat{e}_3)$$

$$c_1 v_1 + c_2 v_2 + c_3 v_3 = \boxed{\vec{c} \cdot \vec{v}} \leftarrow \begin{array}{l} \text{most general} \\ \text{linear function} \end{array}$$

↑
for some
vector $\vec{c}$